Infosys - Coding 12

HackerRank

In the kingdom of Auroria, the once-in-a-century Grand Festival is about to begin. A sacred ritual by the Elders of Balance: The Ritual of Spheres. This ritual requires placing enchanted orbs onto mystical altars arranged in a grand circle. The energy from these orbs fuels the festival's magic.

There are N different types of enchanted orbs, each radiating with a unique magical aura, numbered 1 to N. The i-th type of orb has exactly A[i] orbs available. There are M mystical altars, each with a capacity limit of B[j] meaning an altar can hold no more than B[j] orbs in total.

However, the Elders of Balance have set an ancient restriction:

The j-th altar can hold at most (i × j) orbs of type i, ensuring the equilibrium of magical forces across the festival grounds.

Your duty is to find the maximum total number of orbs that can be placed onto the altars while respecting the sacred balance.

Input Format

The first line contains an integer, N, denoting the number of elements in A.

The next line contains an integer, M, denoting the number of elements in B.

Each line i of the N subsequent lines (where 1 ≤ i ≤ N) contains a long value describing A[i].

Each line i of the M subsequent lines (where 1 ≤ i ≤ M) contains a long value describing B[i]. (this line was cut off at the bottom of image 2, inferred from the pattern)

Constraints

1 <= N <= 500

1 <= M <= 500

1 <= A[i] <= 10^12

1 <= B[i] <= 10^12

Output Format

The output contains an integer value

Sample Input 0

```
1
1
1
2
```

Sample Output 0

```
1
```

Explanation 0

N=1 M=1 A=[1] B=[2] There is only one solution with 1 and 2

Sample Input 1

```
2
2
2
1
1
2
```

Sample Output 1

```
3
```

Explanation 1

N=2 M=2 A=[2 1] B=[1 2] Place 1 orb of type 1 in 1st altar, 2nd orb of type 1 in 2nd altar. Place 1 orb of type 2 in altar 2.